

GenResume

Project Team

Umer Asghar	21I-0409
Mir Mujtaba Habib	21I-0661
Samama Taimoor	21I-0388

Session 2021-2025

Supervised by

Dr. Danish Shehzad

Co-Supervised by

Mr. Aqib Rehman



Department of Computer Science

**National University of Computer and Emerging Sciences
Islamabad, Pakistan**

June, 2025

Table of Contents

1.1	Problem Statement	1
1.2	Scope.....	2
1.3	Modules.....	3
1.3.1	User Profile Creation & Data Management	3
1.3.2	Basic Resume Generation & Job Aggregation.....	3
1.3.3	Advanced Role-Based Insights & Skill Matching	4
1.3.4	Basic Resume Generation & Job Aggregation.....	4
1.4	User Classes and Characteristics.....	5
2.1	High Level Use Cases	6
	Expanded Use Cases	10
2.2	Functional Requirements	19
2.3	Non-Functional Requirements	20
2.3.1	Quality Attributes.....	20
2.4	Domain Model	21
	Chapter 3 System Overview	22
3.1	Architectural Design	22
3.2	Design Models	23
3.3	Data Design.....	28

List of Figures

2.1	Domain Model Diagram	10
3.1	Data Flow Diagram	11
3.2	Use Case Diagram	12
3.3	Class Diagram	12
3.4	SQ-1	13
3.5	SQ-2	13
3.6	SQ-3	13
3.7	SQ-4	14
3.8	SQ-5	14
3.9	SQ-6	14
3.10	SQ-7	15
3.11	SQ-8	15
3.12	SQ-9	15

Chapter 1

Introduction

Both LinkedIn, indeed and Glassdoor are recruitment sites that provide job post/positions and simple resume creation tools but do not have advanced and basic features, to enhance the users, feed-back loop. Most platforms to date are limited in functionality – resume builders are mainly concerned with design and minimal content, while the job boards make their search based on keywords. That is why candidates are not always offered a job they desire because their resumes may not be optimized or the matching has been done improperly.

Numerous products, such as Jobscan and Rezi, make users more ATS-friendly but do not connect job suggestions or career planning, leaving users trapped in a web of non-intersecting tools for a highly interconnected job world. This is most un-strategic in terminals as it makes it hard for applicants in their job search to leverage on their applications the best way possible as they seek the next career step.

To fill these gaps, GenResume covers resume enhancement, job search, skills assessment, and career advice all in one. With features of artificial intelligence, GenResume seeks to be an all-inclusive support service towards job searching.

1.1 Problem Statement

CV creation for the ATS specialists is a major hurdle for the job seekers, the process of job search for the right fit is equally a problem and the need-to-know what skills one lacks for the desired job is another challenge. Currently available tools cover only one or two steps in the job application process for example formatting of resume or, job search but lacks an over- arching solution. This disconnection results in job seekers being unable to enhance application documents, match themselves with demand and manage career transitions.

This way, GenResume provides the ability to create unique, ATS-friendly resumes for the intended job title, recommend jobs from the external databases for the user, and highlight possible gaps in the user's skillset, within the same service.

1.2 Scope

The areas of GenResume include several aspects of the job application process, and its primary goal is to help job seekers find a job with tools that are not available on other platforms. The platform includes:

User Profile Creation and Customization: Users can input all their experience, education, skills, credentials, and projects in an effort to put together a comprehensive profile. This information builds the foundation for creating resumes and getting job recommendations.

ATS-Compliant Resume Generation: It is possible to create an individual resume and, on its basis, develop an ATS-friendly version. Once the candidate picks a particular job title from suggested job ad posting, the system then produces a resume with the right keywords and structures to beat the ATS scans. The specialized platform offers guidance on how well the resume fits the position – including changes that may be made.

Job Search and Personalized Recommendations: The GenResume also links candidates to external job boards such as Indeed and Glassdoor; the results are filtered according to the user's profile. These recommendations are not plain keyword matches; instead, it aligns the user's profile to the job by using AI as a tool for comparison.

Skill Gap Analysis and Learning Resources: The above listed are unique features and by analyzing job descriptions of various positions, the skills and qualifications not matched with the user profile is brought out. It then offers concrete suggestions to use these services such as Coursera and edX to bridge these gaps for users.

Career Path Guidance and Job Market Trends: GenResume is an application that will assist users in mapping out their career progressions as it possesses various real-time job market information. This feature makes user aware of the expectations, emerging positions, and requirements that are valued in a particular field.

1.3 Modules

1.3.1 User Profile Creation & Data Management

The first unit focuses on the prerequisites of constructing and curating user profiles and the data associated with it. This module is designed specially to collect user data and structure it for the sake of other operation.

Features:

1. **Manual Profile Creation:** To provide information about the professional background, users can enter their educational and workplace related data, skills, certifications, and projects.
2. **Data Management System:** Well-organized and easily accessible system of maintaining the data about users with the possibility of their profile updates.
3. **GitHub Integration:** It also supports users to link the GitHub account to import the entire data automatically while creating the profile.
4. **Profile Editing:** Users can update their profiles with new skills, certifications, or experience as they progress in their careers.

1.3.2 Basic Resume Generation & Job Aggregation

- This module builds upon the user profile to enable resume generation and job aggregation functionalities.

Features:

1. **Basic Resume Generation:** Users can generate resumes using predefined templates. These resumes will pull data from the user's profile to create tailored documents.
2. **Multiple ATS-Compliant Templates:** Offering a variety of templates that comply with ATS guidelines to improve the chances of resumes passing through automated systems.
3. **Job Aggregation:** Integration with job boards (e.g., LinkedIn, Indeed, Glassdoor) to aggregate job listings relevant to the user's profile.
4. **Job Search Functionality:** Basic search features that allow users to search for jobs by role, location, or company.

1.3.3 Advanced Role-Based Insights & Skill Matching

- This module introduces advanced role-based insights and skill matching features, enhancing the user's ability to target specific roles and identify areas for improvement.

Features:

1. **Role-Based Career Insights:** Providing users with detailed analysis on the skills, experience, and qualifications needed for specific roles, such as AI/ML Engineer, Data Scientist, etc.
2. **Skill Matching:** Comparing user profiles with job descriptions to identify matching skills and highlight areas where the user may need further development.
3. **Skill Insights:** Providing more descriptive knowledge about a given skill, its applicability, how it is used in the market and the occupations related to it.
4. **Career Path Analysis:** Proposing the actual and possible training and development programs according to the user's existing competencies and target occupation.

1.3.4 Basic Resume Generation & Job Aggregation

- The last one is dedicated to the supplement to the platform and utilizes the integration of AI and NLP to improve core functions of a platform.

Features:

1. **AI-Powered Resume Generation:** Applying natural language processing technology in order to read a job description and create an original resume that corresponds to all the conditions.
2. **NLP-Based Job Description Analysis:** For instance, using NLP to summarize and analyze job descriptions so that the user benefits by finding out what the job really implies.
3. **Advanced Skill Gap Analysis:** An application of artificial intelligence for the purpose of estimating the time and work experience necessary for training according to the profile established by the user and his/her goals.
4. **AI-Driven Job Recommendations:** Matching users with job offers according to the information provided by the users and their submitted applications.

1.4 User Classes and Characteristics

Below are the various user classes that will use this product

User class	Description
Job Seekers	It targets individuals who will use the created GenResume in a search of jobs, get career advice and create their resume.
Recruiters/Employers	Those hiring managers, or companies which will be the indirect beneficiaries receiving highly qualified candidates, whose resumes passed the ATS.
Educational Platforms	Affiliates such as Coursera, edX and so on, that can provide courses that can fill the skill gaps which the system has already discovered.
Universities and Career Centers	Schools that could make use of this tool to enable students and graduates enhance their marketability.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 High Level Use Cases

Use Case ID UC-101

Use Case Name User Registration and Authentication

Actors Job Seeker

Type Primary

Description The user signs up on the GenResume platform using their email, or through a social media account. Once the user goes through the registration process s/he will then log in the system and find tools such as resume builder, job search, and career suggestions.

Use Case ID UC-102

Use Case Name User Profile Creation and Customization

Actors Job Seeker

Type Primary

Description The user completes and adds to their professional account the basics such as personal information, education, skills, work experience, certifications, and projects. It doubles up as the foundation for job suggestions and creating resumes or updating an existing one.

Use Case ID UC-103

Use Case Name Job Search and Recommendation

Actors Job Seeker

Use Case ID UC-103**Type** Primary**Description** External job boards are integrated into the platform where the user can find the list of external job vacancies and receive tailored job suggestions. Job seekers can look for job openings and put in search parameters like location, type of job and many more.

Use Case ID UC-104**Use Case Name** ATS-Compliant Resume Generation**Actors** Job Seeker**Type** Primary**Description** The users create the resumes which are compatible with the Applicant Tracking Systems. The system scans through the descriptions of jobs, matches what the user is seeking, and guarantees inclusion of the proper keywords. Applicant tracking system friendly built resume.

Use Case ID UC-105**Use Case Name** Cover Letter Generation**Actors** Job Seeker**Type** Primary**Description** Depending on the user's characteristics and requested position, the software creates a cover letter that may be used while posting resumes.

Use Case ID UC-106**Use Case Name** Skill Gap Analysis**Actors** Job Seeker**Type** Primary**Description** The platform looks at the user's current profile to match it with job descriptions in order to see possible gaps in their skills. It offers recommendations where income-earning opportunities, online courses or other pertinent resources can be found to close these gaps.

Use Case ID UC-107**Use Case** Career Path Guidance

Use Case ID UC-107**Name****Actors** Job Seeker**Type** Primary**Description** Candidates get advice about career opportunities in light of their profile, current job and trends and categories of jobs. It offers guidance on the promotion in the chosen field, for example, gain new knowledge or certification.

Use Case ID UC-108**Use Case Name**

Job Market Trends Analysis

Actors Job Seeker**Type** Primary**Description** It brings data from job sites and other sources to share information on the current trends in job openings. There is sector, skill and role-based trends that help users in making right career choices.

Use Case ID UC-109**Use Case Name**

Resume and Cover Letter Preview and Download

Actors Job Seeker**Type** Primary**Description** Possibilities of the Resume/Cover Letter Generator After creating resume and cover letters, users are able to view the documents and may edit the details if needed, and the documents can be downloaded in PDF format for employment purposes.

Use Case ID UC-110**Use Case Name**

External API Integration for Job Listings

Actors System**Type** Primary**Description** The platform, for example, can connect with other job websites (organizations such as Indeed or Glassdoor) by using APIs and pull the most up-to-date jobs. The listings are next presented to users based on a match between the users' profiles.

Use Case ID UC-111

Use Case Name Profile and Resume Updates

Actors Job Seeker

Type Primary

Description Users can give fresh content every now and then by adding new encounters in the work experience list, new skills or certifications acquired. The system recalculates job recommendations and reforms the resume.

Expanded Use Cases

Use Case ID: **UC-101**

Use Case Name: User Registration and Authentication

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has internet access and is on the GenResume platform.

Triggers:

- The user navigates to the registration page or tries to access restricted features that require login.

Main Flow:

1. The user clicks on the “Register” button.
2. The system presents a registration form.
3. The user provides their email address or selects a social media login (e.g., Google, LinkedIn).
4. The user creates a password (if email is used) and fills in any other required fields (e.g., name).
5. The system validates the provided information.
6. If validation is successful, the system creates a new account and sends a confirmation email (if email registration is used).
7. The user receives the confirmation email and clicks on the activation link (email users only).
8. The user logs in and is redirected to the dashboard.

Alternative Flows:

- **AF1:** If the user opts for social media login, they are redirected to the provider’s authentication page (e.g., Google) and after authentication, the system creates the account without further steps.

Exception Flows:

- **EF1:** If the user provides an invalid email or password format, the system displays an error and prompts for correction.
- **EF2:** If social media authentication fails, the user is redirected back to the registration page with an error message.

Postconditions:

- The user is successfully registered and authenticated, and can now access platform features.

Use Case ID: UC-102

Use Case Name: User Profile Creation and Customization

Actors: Job Seeker

Type: Primary

Preconditions:

- The user is logged into the platform.

Triggers:

- The user selects the "Create Profile" option from the dashboard.

Main Flow:

1. The system displays a form for entering personal information (e.g., name, contact info).
2. The user fills in education, skills, work experience, and certifications.
3. The user uploads any additional documents (e.g., certificates or project details).
4. The system validates the entered information.
5. The system saves the profile and confirms the profile creation.

Alternative Flows:

- **AF1:** The user skips some non-mandatory fields like uploading documents or adding extra skills.
- **AF2:** The user chooses to import information from their LinkedIn profile (API integration).

Exception Flows:

- **EF1:** If the user leaves mandatory fields empty (e.g., name, education), the system prompts for completion.
- **EF2:** If document upload fails, the system displays an error and allows the user to retry.

Postconditions:

- The user's profile is created and available for job matching and resume generation.
-

Use Case ID: UC-103

Use Case Name: Job Search and Recommendation

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has created a profile with relevant information (e.g., skills, work experience).
-

Triggers:

- The user navigates to the "Job Search" section of the platform or views recommendations on their dashboard.

Main Flow:

1. The system compares the user's profile data (e.g., skills, location) with job listings from external job portals (Indeed, Glassdoor).
2. The system displays personalized job recommendations.
3. The user can manually search for jobs by applying filters like location, job type, and salary range.
4. The user clicks on a job listing to view its details.
5. The user applies for the job by clicking the "Apply" button, which redirects them to the respective job portal.

Alternative Flows:

- **AF1:** The user bookmarks a job to apply for later.
- **AF2:** The user refines the search criteria (e.g., changes location or keyword) and gets updated results.

Exception Flows:

- **EF1:** If there are no matches, the system displays a message suggesting changes to the profile or search criteria.

Postconditions:

- The user has viewed and/or applied to suitable job recommendations based on their profile.

Use Case ID: UC-104

Use Case Name: ATS-Compliant Resume Generation

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has created a profile with all relevant details.

Triggers:

- The user selects the "Generate Resume" option from the dashboard.

Main Flow:

1. The system analyzes the user's profile information (e.g., skills, experience).
2. The user selects an ATS-compliant template.

3. The system optimizes the resume by matching keywords from the user's profile with job descriptions and ATS standards.
4. The system generates a preview of the resume.
5. The user reviews the preview and makes any necessary edits.
6. The system generates the final version of the resume in the chosen format (e.g., PDF).

Alternative Flows:

- **AF1:** The user chooses a different resume template after previewing the initial option.

Exception Flows:

- **EF1:** If the resume generation fails, the system displays an error and suggests trying again later.

Postconditions:

- The user has a fully generated, ATS-compliant resume ready for download or application submission.
-

Use Case ID: UC-105

Use Case Name: Cover Letter Generation

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has created a profile and selected a job to apply for.

Triggers:

- The user selects the "Generate Cover Letter" option for a specific job application.

Main Flow:

1. The system extracts relevant information from the user's profile (e.g., name, skills, experience).
2. The system generates a cover letter tailored to the selected job, including job title, company name, and specific requirements.
3. The user previews the cover letter and makes any necessary changes.
4. The system finalizes the cover letter and makes it available for download or submission.

Alternative Flows:

- **AF1:** The user edits the auto-generated cover letter to include personalized content.

Exception Flows:

- **EF1:** If the cover letter generation fails, the system prompts the user to retry later.

Postconditions:

- The user has a personalized cover letter ready to apply for the job.
-

Use Case ID: UC-106

Use Case Name: Skill Gap Analysis

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has created a profile and selected job preferences.

Triggers:

- The user initiates a skill gap analysis or views job recommendations.

Main Flow:

1. The system compares the user's profile (skills and experience) with the requirements of the desired job.
2. The system identifies any gaps between the user's qualifications and the job requirements.
3. The system suggests online courses, certifications, or other resources to help the user fill these gaps.
4. The user is presented with a list of learning resources (e.g., Coursera, edX).

Alternative Flows:

- **AF1:** The user selects a resource and navigates to the provider's platform to begin a course.

Exception Flows:

- **EF1:** If no skill gaps are found, the system displays a message informing the user that they meet the job requirements.

Postconditions:

- The user is informed about any skill gaps and provided with resources to bridge those gaps.
-

Use Case ID: UC-107

Use Case Name: Career Path Guidance

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has created a complete profile and set career goals.

Triggers:

- The user accesses the "Career Guidance" section from the dashboard.

Main Flow:

1. The system evaluates the user's profile, including current skills, experience, and career goals.
2. The system generates personalized career advice, suggesting next steps such as acquiring new skills or switching industries.
3. The user views a list of suggested career paths, skills to develop, and certifications to pursue.
4. The user selects a recommended path to explore further resources.

Alternative Flows:

- **AF1:** The user adjusts career goals, and the system updates the recommendations.

Exception Flows:

- **EF1:** If no suitable career advice is available, the system suggests expanding or revising the user's profile information.

Postconditions:

- The user has received personalized career guidance, with actionable steps to progress in their career.
-

Use Case ID: **UC-108**

Use Case Name: Job Market Trends Analysis

Actors: Job Seeker

Type: Primary

Preconditions:

- The system has integrated external data sources for job market trends.

Triggers:

- The user accesses the "Job Market Trends" section from the dashboard.

Main Flow:

1. The system retrieves data from job portals and external sources regarding the latest market trends.
2. The system analyzes the data and identifies trends in specific sectors, skills, or roles.
3. The user is presented with visualizations and insights into market trends relevant to their career goals.
4. The user can filter trends by industry, region, or job type.

Alternative Flows:

- **AF1:** The user bookmarks certain trends for future reference.

Exception Flows:

- **EF1:** If there is no data available for the user's selected industry, the system displays a message suggesting the user try another industry or region.

Postconditions:

- The user is informed about current market trends and how they relate to their career.
-

Use Case ID: UC-109

Use Case Name: Resume and Cover Letter Preview and Download

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has generated a resume or cover letter.

Triggers:

- The user clicks on the "Preview" or "Download" button for a generated document.

Main Flow:

1. The system presents a preview of the resume or cover letter.
2. The user reviews the document and makes edits if necessary.
3. The system regenerates the updated resume or cover letter.
4. The user clicks the "Download" button.
5. The system generates a PDF and initiates the download.

Alternative Flows:

- **AF1:** The user changes the template of the resume or cover letter and generates a new preview.

Exception Flows:

- **EF1:** If the download fails, the system retries and provides an error message if it continues to fail.

Postconditions:

- The user has successfully downloaded the final version of their resume or cover letter.
-

Use Case ID: UC-110

Use Case Name: External API Integration for Job Listings

Actors: System

Type: Primary

Preconditions:

- APIs for external job portals (e.g., Indeed, Glassdoor) are available.

Triggers:

- The system initiates a scheduled job to fetch external job listings.

Main Flow:

1. The system sends API requests to external job portals for job listings.
2. The external systems return job listings in a structured format.
3. The system parses the job listings and filters out irrelevant or duplicate entries.
4. The system updates the job listing database with new entries.
5. Job seekers can now view the updated listings.

Alternative Flows:

- **AF1:** The system refreshes listings periodically to ensure the latest job opportunities are available.

Exception Flows:

- **EF1:** If an API call fails, the system retries after a delay. If the error persists, the system logs the error and notifies the administrator.

Postconditions:

- The job listings database is up-to-date with external jobs, ready for user searches.
-

Use Case ID: UC-111

Use Case Name: Profile and Resume Updates

Actors: Job Seeker

Type: Primary

Preconditions:

- The user has already created a profile and generated a resume.

Triggers:

- The user logs into the platform and selects the option to update their profile or resume.

Main Flow:

1. The system displays the user's existing profile, with options to update sections such as work experience, skills, and education.
2. The user modifies or adds new information (e.g., a new job position, recently completed certification).
3. The system validates the updated information and saves the changes.
4. The user chooses to regenerate their resume based on the updated profile.
5. The system regenerates the resume, ensuring that it remains ATS-compliant.
6. The user previews the updated resume and downloads it in the desired format (e.g., PDF).

Alternative Flows:

- **AF1:** The user updates only the profile without regenerating the resume.
- **AF2:** The user chooses to use a different template when regenerating the resume.

Exception Flows:

- **EF1:** If the update fails (e.g., due to incomplete information), the system prompts the user to correct the missing or invalid fields.
- **EF2:** If resume generation fails, the system displays an error and suggests trying again later.

Postconditions:

- The user's profile is successfully updated, and the resume has been regenerated and downloaded, if requested.

2.2 Functional Requirements

1. **User Registration and Authentication:** It enables users to record and sign on with their email or social networking account details.
2. **Profile Creation and Customization:** The it allows users to create and edit profiles personal information and education and work experience, certifications and skills.
3. **ATS-Compliant Resume Generation:** Job seekers are able to write CVs that are optimized to match the descriptions of jobs of interest and gain feedback on the level of readmittance in the system. There are many templates to select from and the users can change the appearance of the resume easily.
4. **Cover Letter Generation:** The cover letters are created automatically from the profile and the description of the vacancy, which users can then modify.
5. **Job Search and Personalized Recommendations:** The jobs are sourced from other job search engines like Indeed and Glassdoor to recommend to a user based on skills and interests. Under the section of jobs, users can search jobs through location, job type, and salary.
6. **Skill Gap Analysis:** Such users' profiles are matched with the job descriptions and the courses that can be useful for filling in the gaps are proposed.
7. **Career Path Guidance:** From the information provided users are in a position to be advised on how best to advance in their career based on the existing job market.
8. **Job Market Trends:** It allows finding out what skills are relevant on the market and in which positions, salary ranges for a certain occupation, and promising sectors.
9. **Resume and Job Matching Score:** The set of features of the resume is rated and an ATS rating and a matching score are returned for the given resume with tips on how to increase them.
10. **User Feedback and Suggestions:** Instant feedback of Resume writing tips, Profile building tips, Job application tips etc.

2.3 Non-Functional Requirements

1. **Performance:** The platform response time should be fast as far as three seconds while for some actions like resume generation and employment recommendation.
2. **Security:** This information contains profiles, resumes, and it has to be safe with the help of the encryption and reliable login options such as 2FA.
3. **Availability:** There should be high platform availability, meaning which is the contrary of downtime or the time when the platform is unavailable.
4. **Usability:** Graphic interface should be simple and the platform should be easily accessible through a computer as well as the cell phone.
5. **Maintainability:** It should also be easy to update and repair based on standard protocols of software construction.
6. **Compliance:** The platform has to adhere to laws such as GDPR on the protection of user's data and should be collected with the user's permission.

2.3.1 Quality Attributes

1. **User Satisfaction:** U The audience should get value in using the platform to create resumes, look for jobs and enhance their profiles.
2. **Accuracy:** Currently, job matching and resume feedback should be precise and assist users with matching to job description.
3. **Flexibility:** It should be easy for users to make changes to the resumes and cover letters depending on every job post.
4. **Security:** There are some specifics that are worth discussing: First, user data have to be kept from the public, and there has to be strong privacy.
5. **Portability:** The platform should be readily navigable and retrievable from any device – whether a desktop or a mobile one.

2.4 Domain Model

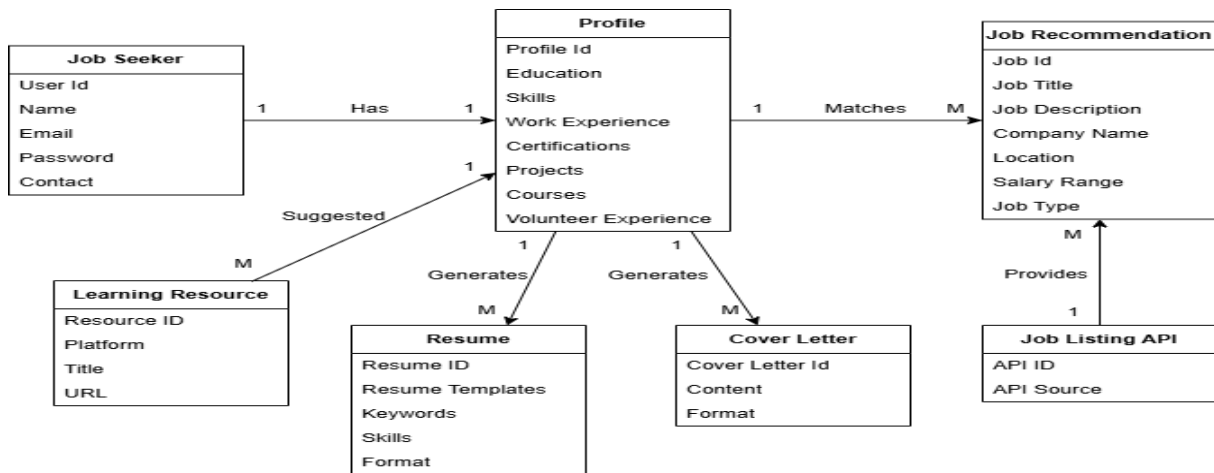


Figure 2.1: Domain Model Diagram

Chapter 3 System

Overview

Give a general description of the functionality, context, and design of your project. Provide any background information if necessary.

3.1 Architectural Design

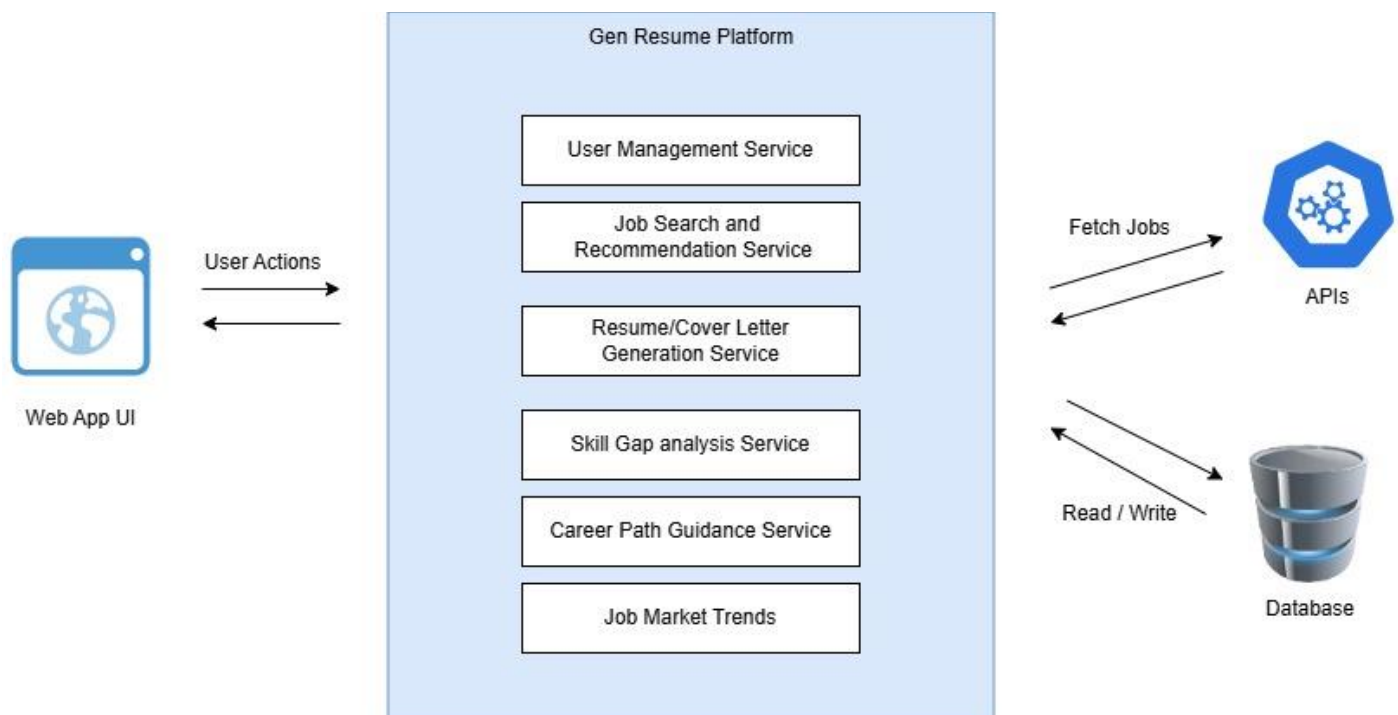


Figure 3.1: Data Flow Diagram

3.2 Design Models

Create design models as are applicable to your system. Provide detailed descriptions with each of the models that you add. Also ensure visibility of all diagrams.

Design Models for Object Oriented Development Approach

The applicable models for the project using object-oriented development approach may include:

Use Case Diagram:

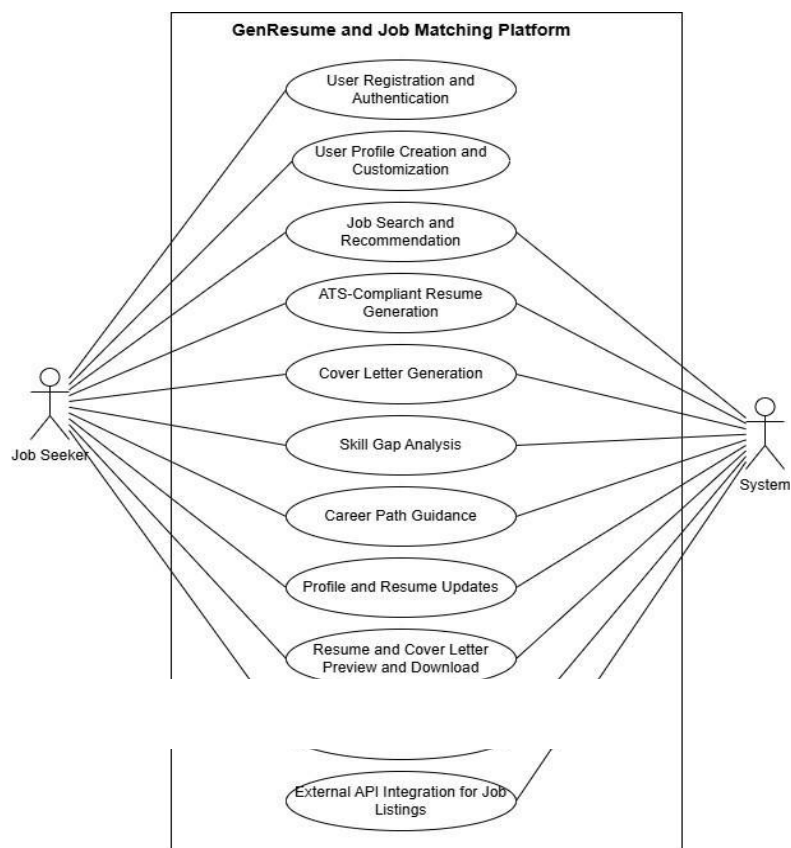


Figure 3.2: Use Case Diagram

• Class Diagram

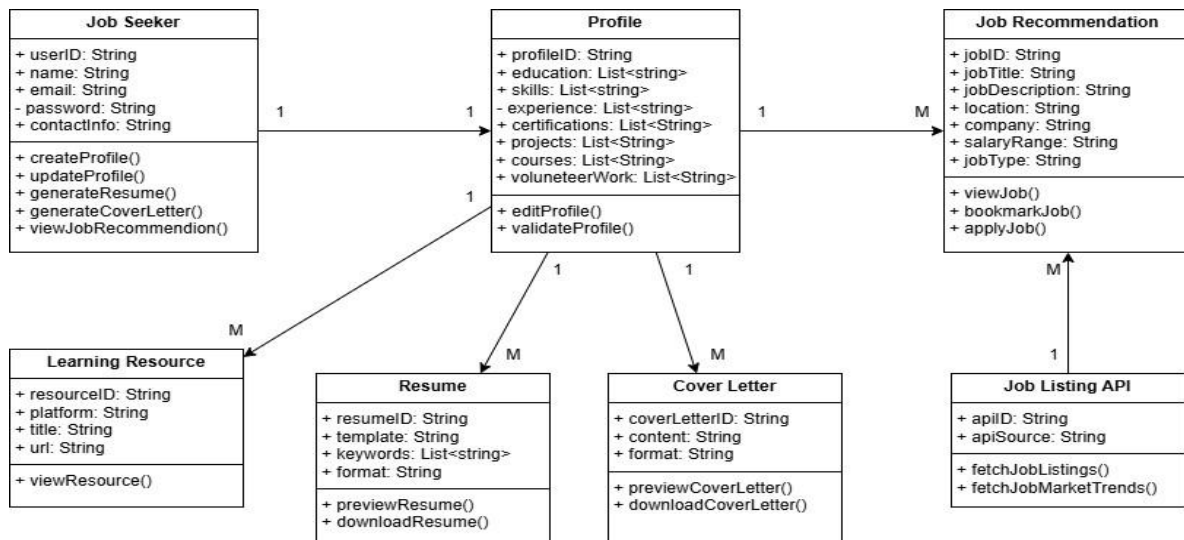


Figure 1

Figure 3.3: Class Diagram

• Class-level Sequence Diagram

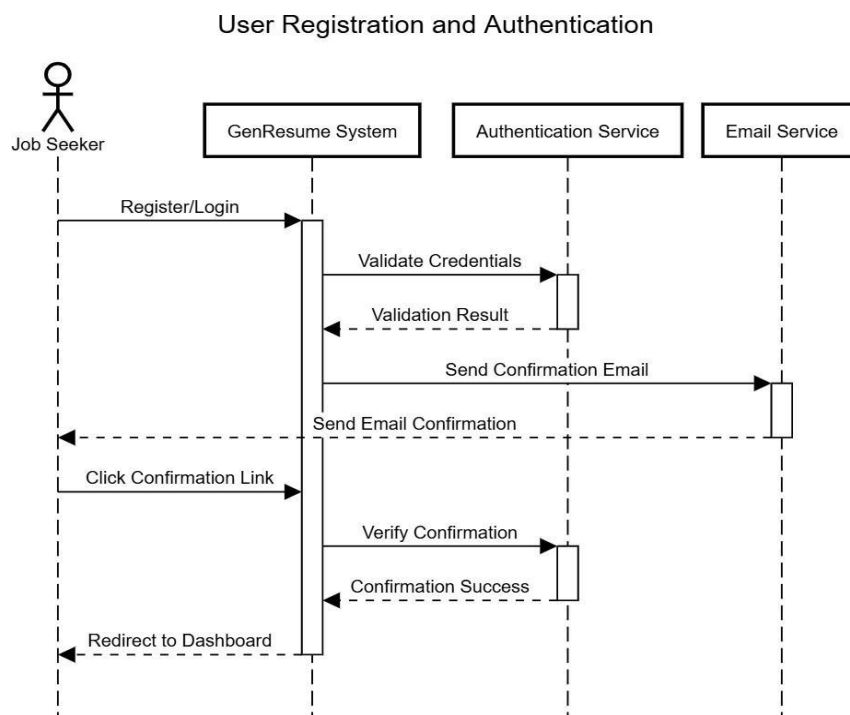


Figure 3.4: SQ-1

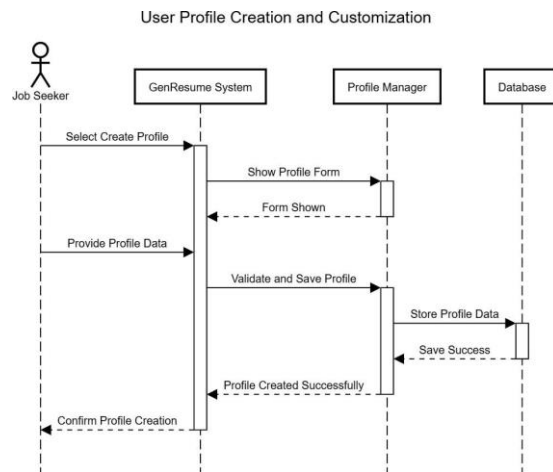


Figure 3.5: SQ-2

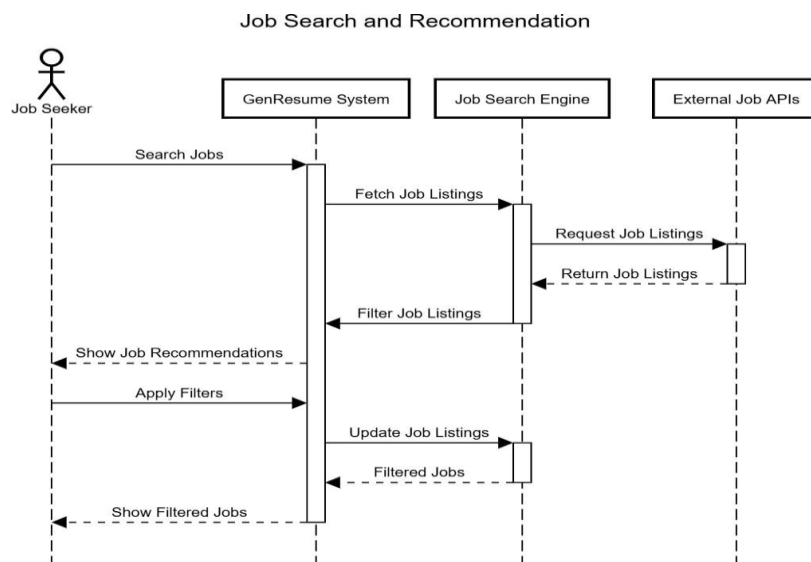


Figure 3.6: SQ-3

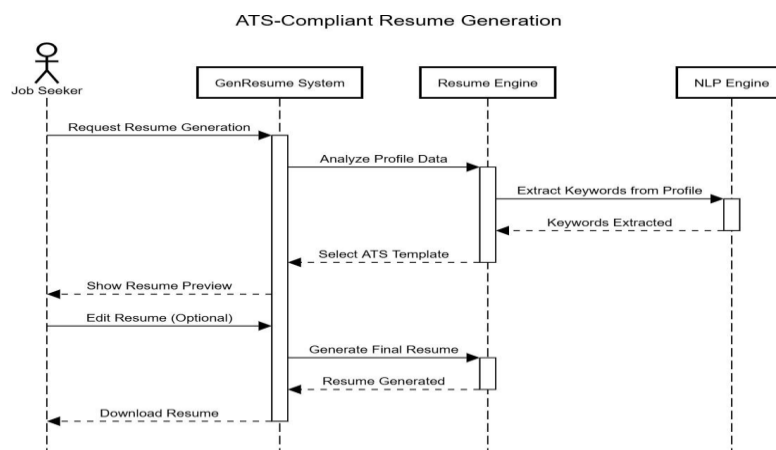


Figure 3.7: SQ-4

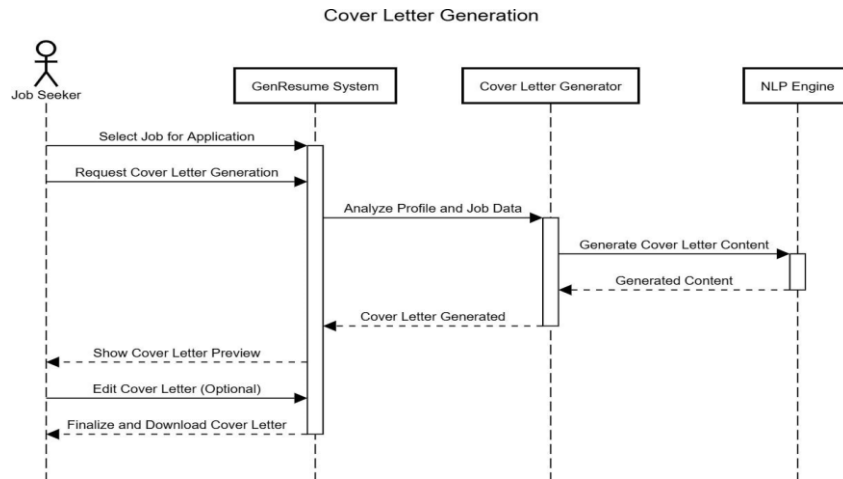


Figure 3.8: SQ-5

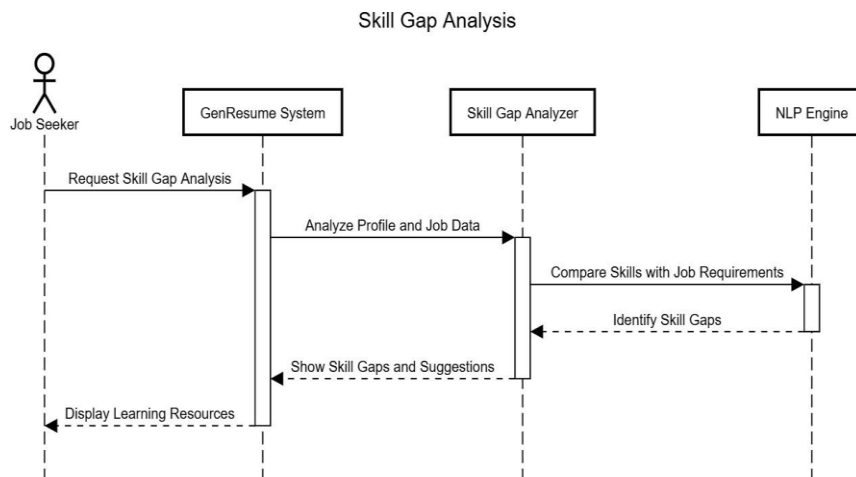


Figure 3.9: SQ-6

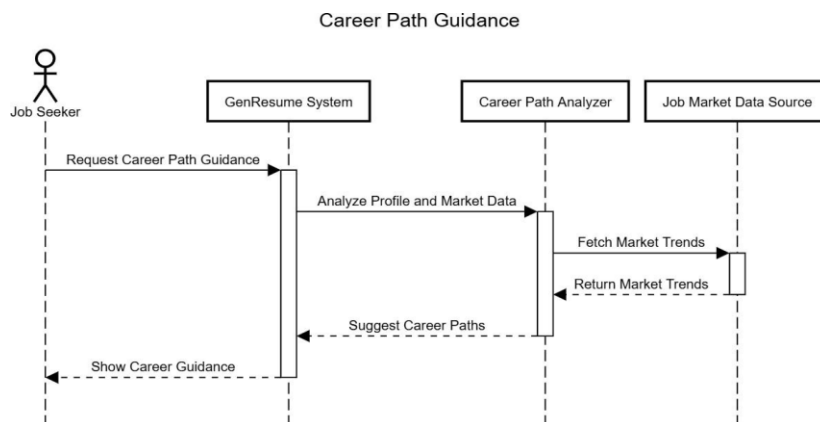


Figure 3.10: SQ-7

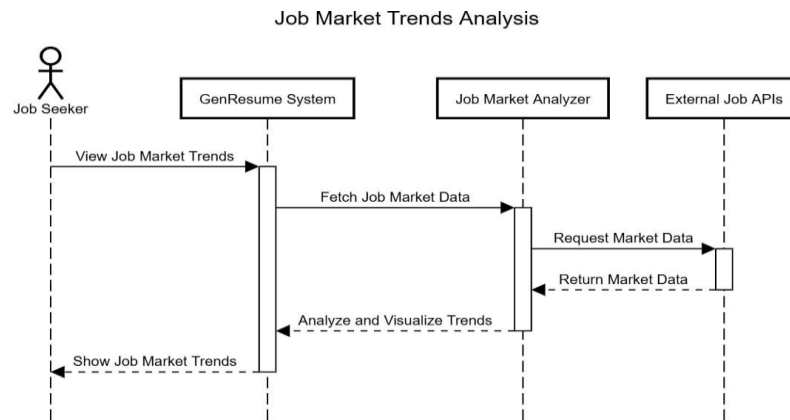


Figure 3.11: SQ-8

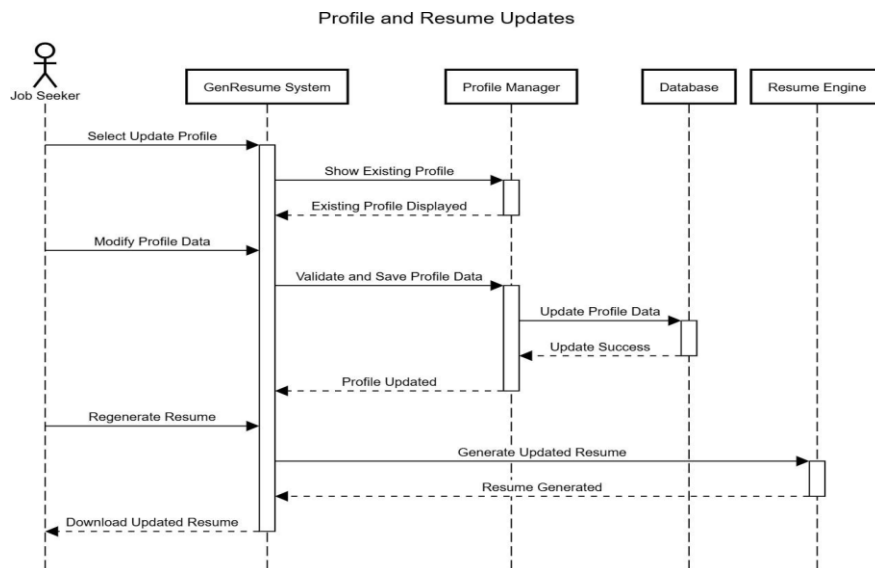


Figure 3.12: SQ-9

3.3 Data Design

Explain how the information domain of your system is transformed into data structures.

Describe how the major data or system entities are stored, processed, and organized.

List any databases or data storage items.

```
"$schema": "http://json-schema.org/draft-07/schema#",
"title": "Job Seeker Application Schema",
"type": "object",
"properties": {
  "JobSeeker": {
    "type": "object",
    "properties": {
      "userID": { "type": "string" },
      "name": { "type": "string" },
      "email": { "type": "string" },
      "password": { "type": "string" },
      "contactInfo": { "type": "string" },
      "profile": { "$ref": "#/definitions/Profile" },
      "resume": { "$ref": "#/definitions/Resume" },
      "coverLetter": { "$ref": "#/definitions/CoverLetter" },
      "jobRecommendations": {
        "type": "array",
        "items": { "$ref": "#/definitions/JobRecommendation" }
      },
      "learningResources": {
        "type": "array",
        "items": { "$ref": "#/definitions/LearningResource" }
      }
    },
    "required": ["userID", "name", "email", "password", "profile"]
  },
  "Profile": {
    "type": "object",
    "properties": {
      "profileID": { "type": "string" },
      "education": { "type": "array", "items": { "type": "string" } },
      "skills": { "type": "array", "items": { "type": "string" } },
      "experience": { "type": "array", "items": { "type": "string" } },
      "certifications": { "type": "array", "items": { "type": "string" } },
      "projects": { "type": "array", "items": { "type": "string" } },
      "courses": { "type": "array", "items": { "type": "string" } },
      "volunteerWork": { "type": "array", "items": { "type": "string" } }
    },
    "required": ["profileID"]
  }
},
```

```
},
"Resume": {
  "type": "object",
  "properties": {
    "resumeID": { "type": "string" },
    "template": { "type": "string" },
    "keywords": { "type": "array", "items": { "type": "string" } },
    "format": { "type": "string" }
  },
  "required": ["resumeID", "template", "keywords", "format"]
},
"CoverLetter": {
  "type": "object",
  "properties": {
    "coverLetterID": { "type": "string" },
    "content": { "type": "string" },
    "format": { "type": "string" }
  },
  "required": ["coverLetterID", "content", "format"]
},
"JobRecommendation": {
  "type": "object",
  "properties": {
    "jobID": { "type": "string" },
    "jobTitle": { "type": "string" },
    "jobDescription": { "type": "string" },
    "location": { "type": "string" },
    "company": { "type": "string" },
    "salaryRange": { "type": "string" },
    "jobType": { "type": "string" }
  },
  "required": ["jobID", "jobTitle", "company", "location"]
}
```

```

},
"LearningResource": {
  "type": "object",
  "properties": {
    "resourceID": { "type": "string" },
    "platform": { "type": "string" },
    "title": { "type": "string" },
    "url": { "type": "string" }
  },
  "required": ["resourceID", "platform", "title", "url"]
},
"JobListingAPI": {
  "type": "object",
  "properties": {
    "apiID": { "type": "string" },
    "apiSource": { "type": "string" },
    "jobListings": {
      "type": "array",
      "items": { "$ref": "#/definitions/JobRecommendation" }
    },
    "jobMarketTrends": {
      "type": "array",
      "items": { "type": "string" }
    }
  },
  "required": ["apiID", "apiSource"]
}

```

```

},
"definitions": {
  "Profile": {
    "type": "object",
    "properties": {
      "profileID": { "type": "string" },
      "education": { "type": "array", "items": { "type": "string" } },
      "skills": { "type": "array", "items": { "type": "string" } },
      "experience": { "type": "array", "items": { "type": "string" } },
      "certifications": { "type": "array", "items": { "type": "string" } },
      "projects": { "type": "array", "items": { "type": "string" } },
      "courses": { "type": "array", "items": { "type": "string" } },
      "volunteerWork": { "type": "array", "items": { "type": "string" } }
    }
  },
  "Resume": {
    "type": "object",
    "properties": {
      "resumeID": { "type": "string" },
      "template": { "type": "string" },
      "keywords": { "type": "array", "items": { "type": "string" } },
      "format": { "type": "string" }
    }
  },
  "CoverLetter": {
    "type": "object",
    "properties": {
      "coverLetterID": { "type": "string" },
      "content": { "type": "string" },
      "format": { "type": "string" }
    }
  }
}

```

```

    },
    "JobRecommendation": {
      "type": "object",
      "properties": {
        "jobID": { "type": "string" },
        "jobTitle": { "type": "string" },
        "jobDescription": { "type": "string" },
        "location": { "type": "string" },
        "company": { "type": "string" },
        "salaryRange": { "type": "string" },
        "jobType": { "type": "string" }
      }
    },
    "LearningResource": {
      "type": "object",
      "properties": {
        "resourceID": { "type": "string" },
        "platform": { "type": "string" },
        "title": { "type": "string" },
        "url": { "type": "string" }
      }
    },
    "JobListingAPI": {
      "type": "object",
      "properties": {
        "apiID": { "type": "string" },
        "apiSource": { "type": "string" },
        "jobListings": {
          "type": "array",
          "items": { "$ref": "#/definitions/JobRecommendation" }
        },
        "jobMarketTrends": {
          "type": "array",
          "items": { "type": "string" }
        }
      }
    }
  }
}

```